



***GPS-BASED ACCIDENT- PRONE  
ZONE IDENTIFICATION AND DRIVER  
ALERT SYSTEM***



**A PROJECT REPORT**

*Submitted by*

|                      |                       |
|----------------------|-----------------------|
| <b>MANIKANDAN K</b>  | <b>(611223114056)</b> |
| <b>MOHAN G</b>       | <b>(611223114060)</b> |
| <b>NAVEENKUMAR R</b> | <b>(611223114066)</b> |
| <b>SABARI K</b>      | <b>(611223114080)</b> |

*In partial fulfillment for the award of the degree*

*of*

**BACHELOR OF ENGINEERING**

**IN**

**MECHANICAL ENGINEERING**

**KNOWLEDGE INSTITUTE OF TECHNOLOGY, SALEM  
(An Autonomous Institution)**

**ANNA UNIVERSITY: CHENNAI 600 02**

**JUNE 2026**



***GPS-BASED ACCIDENT- PRONE  
ZONE IDENTIFICATION AND DRIVER  
ALERT SYSTEM***



**A PROJECT REPORT**

*Submitted by*

|                      |                       |
|----------------------|-----------------------|
| <b>MANIKANDAN K</b>  | <b>(611223114056)</b> |
| <b>MOHAN G</b>       | <b>(611223114060)</b> |
| <b>NAVEENKUMAR R</b> | <b>(611223114066)</b> |
| <b>SABARI K</b>      | <b>(611223114080)</b> |

*In partial fulfillment for the award of the degree*

*of*

**BACHELOR OF ENGINEERING**

**IN**

**MECHANICAL ENGINEERING**

**KNOWLEDGE INSTITUTE OF TECHNOLOGY, SALEM  
(An Autonomous Institution)**

**ANNA UNIVERSITY: CHENNAI 600 025**

**JUNE 2026**

## **BONAFIDE CERTIFICATE**

Certified that this project report “**GPS-BASED ACCIDENT-PRONE ZONE IDENTIFICATION AND DRIVER ALERT SYSTEM**” is the bonafide work of “**MANIKANDAN K, MOHAN G, NAVEENKUMAR R, SABARI K**” who carried out the project work under my supervision.

### **SIGNATURE**

Dr. K.S.PRABHAKARAN, M.E., Ph.D., MISTE,

### **HEAD OF THE DEPARTMENT**

Associate Professor

Department of Mechanical Engineering

Knowledge Institute of Technology

Salem – 637 504.

### **SIGNATURE**

Mr. S.SANTHOSH, B.E., M.E.

### **SUPERVISOR**

Assistant Professor

Department of Mechanical Engineering

Knowledge Institute of Technology

Salem – 637 504.

## ACKNOWLEDGEMENT

I am truly grateful to **Dr.PSS. SRINIVASAN, M.Tech. (IIT-B), Ph.D., FIE**, the Founder & President of KIOT Trust and Executive Chairman of the KIOT, forever thankful to **Dr.J.KUMAR,M.S. (IIT-M), Ph.D.**, Secretary, **Thiru.V.SURESHKUMAR, B.E.**, Treasurer and esteemed members of the KIOT Trust. Their unwavering support and provision of excellent facilities have been instrumental in the successful completion of my project work.

My heartfelt gratitude to **Dr.K. VISAGAVEL, M.E., Ph.D., FIE**, Principal of KIOT, for his invaluable guidance and encouragement throughout the journey.

I am very much obliged to **Dr.K.S. PRABHAKARAN, M.E., Ph.D.**, Associate Professor and Head of the Department, Mechanical Engineering, for his valuable insights that significantly enhanced the quality of my project work.

I am humbled and thankful to my Class Advisor, **Mr.S.SANTHOSH, M.E.**, and Project Work Coordinator **Mr.K.MUGUNDHAN, M.E., (Ph.D.)**, Assistant Professor, Department of Mechanical Engineering, for his continuous motivation and conducting project reviews and providing support to the completion of the project work.

A sincere thanks to my Project Work Supervisor, **Mr.S.SANTHOSH, M.E.**, Assistant Professor, Department of Mechanical Engineering, for his exceptional guidance and unwavering support throughout the project work duration.

I also wish to acknowledge the entire teaching faculty, supporting staffs of the Mechanical Engineering Department, and my fellow students for their assistance and cooperation, which played a vital role in bringing the project work to fruition.

I am forever thankful to all who contributed for the successful completion of the project.

## **ABSTRACT**

The main objective of this project is to develop a GPS-based Accident-Prone Zone Identification and Prevention System to improve road safety by providing real-time warnings to drivers when they approach accident-prone locations. The system continuously tracks the vehicle's position using GPS technology and monitors its speed to identify unsafe driving conditions in hazardous zones. When a vehicle enters a predefined danger zone and exceeds the safe speed limit, the system generates audio and visual alerts to warn the driver and encourage immediate corrective action. By enhancing driver awareness, reducing over-speeding, and providing timely warnings, the proposed system aims to minimize road accidents, improve transportation safety, and contribute to the development of intelligent road safety and smart transportation systems.

**Keywords:** Accident Prone Zone, GPS Tracking, ESP32, Road Safety, Speed Monitoring, Driver Alert System, Accident Prevention, Real-Time Monitoring, Vehicle Safety, Intelligent Transportation System.

## **TABLE OF CONTENTS**

| <b>CHAPTER<br/>NO.</b> | <b>TITLE</b>                                 | <b>PAGE NO.</b> |
|------------------------|----------------------------------------------|-----------------|
|                        | <b>ABSTRACT</b>                              | i               |
|                        | <b>LIST OF FIGURES</b>                       | vi              |
|                        | <b>LIST OF TABLES</b>                        | vii             |
|                        | <b>LIST OF ABBREVIATIONS</b>                 | viii            |
| <b>1</b>               | <b>INTRODUCTION</b>                          | 1               |
|                        | 1.1 BACKGROUND OF ROAD<br>SAFETY             | 1               |
|                        | 1.2 ACCIDENT-PRONE ZONES                     | 1               |
|                        | 1.3 CHALLENGES IN EXISTING SAFETY<br>METHODS | 2               |
|                        | 1.4 NEED FOR INTELLIGENT WARNING<br>SYSTEMS  | 2               |
|                        | 1.5 PROPOSED SYSTEM OVERVIEW                 | 2               |
|                        | 1.6 SCOPE OF THE PROJECT                     | 3               |
|                        | 1.7 OBJECTIVES OF THE PROJECT                | 3               |
|                        | 1.8 NEED FOR THE PROPOSED SYSTEM             | 4               |
|                        | 1.9 ORGANIZATION OF THE REPORT               | 4               |
| <b>2</b>               | <b>LITERATURE REVIEW</b>                     | 5               |
| <b>3</b>               | <b>PROBLEM IDENTIFICATION</b>                | 8               |
|                        | 3.1 PROBLEM IDENTIFIED                       | 8               |

|                                                 |           |
|-------------------------------------------------|-----------|
| 3.1.1 Delayed Detection of Accident-Prone Zone  | 8         |
| 3.1.2 Over-speeding in Hazardous Zones          | 8         |
| 3.1.3 Ineffective Conventional Warning System   | 9         |
| 3.1.4 Delayed Driver Response                   | 9         |
| 3.1.5 Absence of Real-Time Alerts               | 10        |
| 3.1.6 Increased Accident Risk in Critical Areas | 10        |
| 3.1.7 Need for Intelligent Road Safety Systems  | 10        |
| 3.2 PROPOSED SOLUTION                           | 11        |
| 3.3 OBJECTIVES OF THE PROPOSED SYSTEM           | 12        |
| 3.3.1 Identification of Accident-Prone Zones    | 12        |
| 3.3.2 Driver Alert System                       | 12        |
| 3.3.3 Speed Monitoring                          | 12        |
| 3.3.4 Real-Time Location Tracking               | 12        |
| 3.3.5 Improvement of Road Safety                | 12        |
| 3.3.6 Cost-Effective Implementation             | 12        |
| 3.3.7 Future Scope and Expandability            | 13        |
| 3.3.8 Reduction of Accident Risks               | 13        |
| <b>4 METHODOLOGY</b>                            | <b>14</b> |

|          |                                                     |           |
|----------|-----------------------------------------------------|-----------|
| <b>5</b> | <b>HARDWARE AND COMPONENT SELECTION</b>             | <b>18</b> |
|          | 5.1 ESP32 MICROCONTROLLER                           | 18        |
|          | 5.2 GPS MODULE (NEO-6M)                             | 19        |
|          | 5.3 BUZZER                                          | 20        |
|          | 5.4 LED INDICATOR                                   | 20        |
|          | 5.5 POTENTIOMETER (SPEED SIMULATION UNIT)           | 21        |
|          | 5.6 POWER SUPPLY UNIT                               | 22        |
|          | 5.7 LCD DISPLAY                                     | 22        |
| <b>6</b> | <b>COST ESTIMATION</b>                              | <b>23</b> |
| <b>7</b> | <b>SYSTEM IMPLEMENTATION, OPERATION AND TESTING</b> | <b>24</b> |
|          | 7.1 WORKING                                         | 25        |
|          | 7.1.1 GPS Data Acquisition                          | 26        |
|          | 7.1.2 Accident-Prone Zone Detection                 | 26        |
|          | 7.1.3 Driver Alert Generation                       | 26        |
|          | 7.1.4 Speed Monitoring and Warning                  | 26        |
|          | 7.1.5 Driver Alert Mechanism                        | 27        |
|          | 7.1.6 Improvement of Road Safety                    | 27        |
|          | 7.1.7 System Reliability                            | 27        |
|          | 7.1.8 Overall System Performance                    | 27        |
|          | 7.1.9 Results and Validation                        | 27        |
| <b>8</b> | <b>CONCLUSION</b>                                   | <b>29</b> |



|           |                                                      |           |
|-----------|------------------------------------------------------|-----------|
| <b>9</b>  | <b>ADVANTAGES, APPLICATIONS AND<br/>FUTURE SCOPE</b> | <b>30</b> |
|           | 9.1 ADVANTAGES                                       | 30        |
|           | 9.2 APPLICATIONS                                     | 31        |
|           | 9.3LIMITATIONS                                       | 32        |
|           | 9.4FUTURE SCOPE                                      | 32        |
| <b>10</b> | <b>APPENXDICES</b>                                   | <b>34</b> |
|           | 10.1 APPENDIX – 1 (PHOTOGRAPH)                       | 34        |
|           | 10.2 APPENDIX – 2 (CODE)                             | 34        |
|           | <b>REFERENCES</b>                                    | <b>43</b> |

## LIST OF FIGURES

| <b>FIGURE<br/>NO.</b> | <b>DESCRIPTION</b>                    | <b>PAGE<br/>NO.</b> |
|-----------------------|---------------------------------------|---------------------|
| 4.1                   | Block Diagram                         | 15                  |
| 4.2                   | Flowchart                             | 17                  |
| 5.1                   | ESP 32 Microcontroller                | 18                  |
| 5.2                   | GPS Module (NEO-6M)                   | 19                  |
| 5.3                   | Buzzer                                | 20                  |
| 5.4                   | LED Indicator                         | 20                  |
| 5.5                   | Potentiometer (Speed Simulation Unit) | 21                  |
| 5.6                   | Battery                               | 22                  |
| 7.1                   | System Operation                      | 24                  |
| 7.2                   | Prototype Testing                     | 25                  |
| 7.3                   | Speed Monitoring System               | 28                  |
| 8.1                   | Prototype Output                      | 29                  |
| 10.1                  | Prototype Photograph                  | 34                  |

## **LIST OF TABLES**

| <b>TABLE NO</b> | <b>DESCRIPTION</b> | <b>PAGE NO</b> |
|-----------------|--------------------|----------------|
| 6.1             | Cost Estimation    | 23             |

## LIST OF ABBREVIATIONS

|              |                                                        |
|--------------|--------------------------------------------------------|
| <b>UNO</b>   | Arduino Uno Microcontroller Board                      |
| <b>ESP32</b> | Embedded System with Wi-Fi & Bluetooth Microcontroller |
| <b>IoT</b>   | Internet of Things                                     |
| <b>UNO</b>   | Universal Numbering Object                             |
| <b>IR</b>    | Infrared Radiation                                     |
| <b>GSM</b>   | Global System for Mobile communication                 |
| <b>MOB</b>   | Man Overboard                                          |
| <b>GPS</b>   | Global Positioning System                              |
| <b>DC</b>    | Direct Current                                         |

# **CHAPTER 1**

## **INTRODUCTION**

### **1.1 BACKGROUND OF ROAD SAFETY**

Road transportation plays a vital role in modern society by enabling the movement of people and goods. However, the increasing number of vehicles on roads has resulted in a significant rise in road accidents. Many accidents occur in accident-prone zones such as sharp curves, school zones, hospital areas, highways, and construction sites. These accidents are often caused by over-speeding, lack of driver awareness, and inadequate warning systems. Therefore, advanced safety systems are required to reduce accident risks and improve road safety

### **1.2 ACCIDENT-PRONE ZONES**

Many accidents in these areas are caused by over-speeding, driver negligence, lack of awareness, and failure to follow traffic regulations. Drivers who are unfamiliar with a route may not recognize the presence of a hazardous zone until it is too late to react safely. In addition, adverse weather conditions such as rain, fog, and poor lighting can further increase the risk of accidents in these locations.

Many accidents in these zones are caused by over-speeding, lack of driver awareness, and failure to identify hazardous road conditions in advance. Poor weather conditions such as rain and fog can further increase the risk of accidents.

Therefore, it is important to identify accident-prone zones and provide timely warnings to drivers. GPS-based alert systems can help drivers reduce speed and drive safely, thereby improving overall road safety.

### **1.3 CHALLENGES IN EXISTING SAFETY METHODS**

Existing road safety methods mainly rely on traffic signboards, warning boards, and speed limit indicators to alert drivers about hazardous locations. However, these methods depend entirely on the driver's attention and response.

In many situations, drivers may overlook warning signs due to distraction, fatigue, poor visibility, or unfamiliarity with the route. As a result, accidents can still occur even when safety signs are present.

### **1.4 NEEDS FOR INTELLIGENT WARNING SYSTEMS**

With the increasing number of road accidents, there is a growing need for intelligent warning systems that can provide real-time safety alerts to drivers. Traditional warning methods alone are often insufficient to prevent accidents in hazardous locations.

Modern technologies such as GPS, sensors, and microcontrollers can be used to automatically detect accident-prone zones and alert drivers when necessary. These systems help improve driver awareness and encourage safer driving behavior. By providing timely warnings and speed monitoring, intelligent warning systems can significantly reduce accident risks and enhance overall road safety.

### **1.5 PROPOSED SYSTEM OVERVIEW**

The proposed Accident Prone Zone Identification and Prevention System is designed to improve road safety by identifying hazardous locations using GPS technology. The system continuously tracks the vehicle's location and compares it with predefined accident-prone zone coordinates. When the vehicle enters a danger zone, the system monitors its speed in real time. If the vehicle exceeds the safe speed limit, visual, audio alerts are generated using an LED and buzzer to warn the driver.

## **1.6 SCOPE OF THE PROJECT**

The scope of this project is to develop a GPS-based Accident Prone Zone Identification System that helps improve road safety by providing timely warnings to drivers when approaching hazardous locations. The system identifies accident-prone zones using predefined GPS coordinates and continuously monitors the vehicle's location in real time.

When a vehicle enters a dangerous zone, the system generates visual and audio alerts to increase driver awareness. It also monitors vehicle speed within these zones and warns the driver if the speed exceeds the safe limit. The proposed system can be integrated into various vehicles and serves as a cost-effective solution for reducing accident risks and enhancing transportation safety.

## **1.7 OBJECTIVES OF THE PROJECT**

The main objective of this project is to develop an Accident Prone Zone Identification System that improves road safety by providing timely warnings to drivers. The system uses GPS technology and an ESP32 microcontroller to identify accident-prone zones based on predefined coordinates.

The project aims to alert drivers through visual and audio indications when they enter a hazardous area. It also monitors vehicle speed within accident-prone zones and generates additional warnings if the speed exceeds the safe limit. By improving driver awareness and encouraging safe driving practices, the system helps reduce the risk of road accidents and enhances overall transportation safety.

## **1.8 NEED FOR THE PROPOSED SYSTEM**

Road accidents are increasing rapidly due to over-speeding, lack of driver awareness, and delayed warnings in accident-prone zones. Traditional safety measures such as signboards and speed breakers are often insufficient, especially during night-time and poor weather conditions. As a result, drivers may fail to recognize hazardous areas and take preventive action in time.

Therefore, there is a need for an intelligent system that can identify accident-prone zones in real time and provide immediate warnings to drivers. The proposed Accident Prone Zone Identification System addresses this need by using GPS technology and speed monitoring to enhance driver awareness, promote safe driving practices, and reduce the risk of accidents.

## **1.9 ORGANIZATION OF THE REPORT**

The report is organized into chapters covering the introduction, literature review, system analysis, and methodology for identifying accident-prone zones. It then presents the system design, implementation, data analysis, results, and discussion of the proposed system. Finally, the report includes the conclusion, future enhancements, references, and appendices.



## **CHAPTER 2**

### **LITERATURE REVIEW**

Various studies have been carried out to improve road safety using technologies such as GPS, embedded systems, and intelligent warning systems. Existing research focuses on accident detection, speed monitoring, and vehicle tracking to reduce accident risks and enhance driver awareness. However, many systems provide only accident detection or speed monitoring separately. Therefore, an integrated system that combines accident-prone zone identification with speed monitoring is required. The proposed system utilizes GPS and ESP32 technology to provide timely alerts and improve road safety.

#### **1. GPS-Based Accident Alert System**

Paper Title: GPS and GSM Based Accident Detection and Alert System

Authors: S. K. Prasad, P. Mahalakshmi, A. John

In this paper, the authors proposed a system that uses GPS and GSM modules to detect vehicle accidents and send alert messages to emergency contacts. The system focuses mainly on post-accident response rather than accident prevention. Although effective in emergency handling, the system does not provide prior warnings in accident-prone zones, which is a major limitation.

#### **2. Intelligent Speed Control System Using GPS**

Paper Title: Intelligent Speed Control System Using GPS for Accident Prevention

Authors: R. Kumar, M. Singh

This research work presents a GPS-based speed control system where vehicle speed is automatically controlled based on road conditions. The system reduces speed in sensitive zones such as schools and hospitals. However, automatic speed control may not be suitable for all real-time driving conditions, and the system lacks driver alert flexibility.

### **3. IoT-Based Vehicle Monitoring System**

Paper Title: IoT Based Smart Vehicle Monitoring and Accident Prevention System

Authors: A. Sharma, V. Patel

The authors developed an IoT-based system for vehicle tracking and monitoring using cloud connectivity. The system provides real-time vehicle status updates and alerts. While the system offers advanced monitoring, it increases complexity and cost due to continuous internet dependency.

### **4. Accident Prone Zone Identification Using GPS**

Paper Title: Accident Prone Zone Detection Using GPS and Embedded System

Authors: K. Ramesh, S. Anitha

This paper focuses on identifying accident-prone zones using predefined GPS coordinates and providing alerts to drivers. The system effectively identifies danger zones but does not consider vehicle speed while issuing warnings. As a result, it may generate unnecessary alerts even when the vehicle is moving at a safe speed.

### **5. Speed Monitoring Using Embedded Systems**

Paper Title: Vehicle Speed Monitoring and Warning System Using Microcontroller

Authors: P. N. Rao, R. Karthik

In this research, a microcontroller-based speed monitoring system is proposed to warn drivers when speed exceeds a predefined limit. The system improves speed compliance but operates independently of location data, making it less effective in location-specific accident prevention.

## **6. Research Gap and Motivation**

From the literature review, it is observed that most existing systems focus either on accident detection after occurrence or speed monitoring without considering location-based risk zones. Very few systems integrate real-time GPS-based accident-prone zone identification with speed monitoring and driver alerts. The proposed system addresses this gap by combining location tracking, speed evaluation, and intelligent alert generation using an ESP32 microcontroller, making it more effective for real-time accident prevention.

## **CHAPTER 3**

### **PROBLEM IDENTIFICATION**

#### **3.1 PROBLEM IDENTIFIED**

Road accidents are increasing due to over-speeding, lack of driver awareness, and insufficient warnings in accident-prone zones. Locations such as sharp curves, school areas, hospital zones, highways, and construction sites are more vulnerable to accidents. Existing safety measures like signboards and speed breakers may not always provide timely warnings to drivers. Therefore, there is a need for a smart system that can identify hazardous zones and alert drivers in real time to improve road safety and reduce accident risks.

##### **3.1.1 Delayed Detection of Accident-Prone Zones**

Many road accidents occur because drivers are not aware of accident-prone zones. Hazardous locations such as sharp curves, school zones, and hospital areas require extra attention while driving. In many cases, drivers fail to recognize these dangerous locations in advance. This may result in delayed reactions and unsafe driving behavior. Improving driver awareness through timely warnings can help drivers take necessary precautions. This reduces accident risks and enhances overall road safety.

##### **3.1.2 Over-speeding in Hazardous Zones**

Over-speeding is one of the leading causes of accidents in accident-prone areas. Drivers often travel at high speeds without considering road conditions and potential hazards.

In locations such as sharp curves, school zones, and construction areas, excessive speed reduces the driver's ability to react quickly. This increases the chances of collisions and loss of vehicle control. Therefore, monitoring vehicle speed and providing timely warnings are essential to ensure safe driving and reduce accident

risks.

### **3.1.3 Ineffective Conventional Warning System**

Conventional warning systems such as signboards and speed breakers are commonly used to alert drivers about hazardous zones. However, these methods may not always be effective in preventing accidents.

During night-time, poor weather conditions, or heavy traffic, drivers may fail to notice these warnings. As a result, they may enter accident-prone areas without taking necessary precautions. Therefore, a real-time alert system is required to provide timely warnings and improve road safety.

### **3.1.4 Delayed Driver Response**

Delayed driver response is another major factor contributing to road accidents. When drivers fail to recognize hazards quickly, they may not have enough time to react appropriately.

The increasing number of road accidents highlights the need for intelligent road safety systems. Traditional safety measures alone are not sufficient to prevent accidents in hazardous zones.

This problem is more common in accident-prone zones such as sharp curves, busy intersections, and construction areas. A delayed response can lead to sudden braking, loss of vehicle control, or collisions. Providing early warnings to drivers can improve reaction time and help to take preventive actions, thereby reducing the risk of accidents.

### **3.1.5 Absence of Real-Time Alerts**

Many existing road safety systems do not provide real-time alerts to drivers. As a result, drivers may not receive timely information about accident-prone zones.

Without immediate warnings, the chances of delayed action and unsafe driving increase. This can lead to accidents, especially in hazardous areas. A real-time alert system helps drivers stay informed and take necessary precautions to ensure safer travel.

### **3.1.6 Increased Accident Risk in Critical Areas**

Accident-prone zones such as sharp curves, school areas, highways, and construction sites have a higher risk of accidents. These locations require greater attention from drivers.

Lack of awareness and excessive speed in such areas can lead to serious accidents. The risk becomes even higher during night-time and adverse weather conditions. Therefore, identifying critical zones and providing timely alerts can help reduce accident risks and improve road safety.

### **3.1.7 Need for Intelligent Road Safety Systems**

The increasing number of road accidents highlights the need for intelligent road safety systems. Traditional safety measures alone are not sufficient to prevent accidents in hazardous zones.

Advanced technologies such as GPS, sensors, and embedded systems can provide real-time monitoring and alerts to drivers. These systems help improve awareness and encourage safer driving practices. Therefore, an intelligent road safety system is essential for reducing accident risks and enhancing overall transportation safety.

### **3.2 PROPOSED SOLUTION**

To address the issues associated with accident-prone zones, a GPS-based Accident Prone Zone Identification System is proposed. The system is designed to identify hazardous road locations and provide timely warnings to drivers before they enter such areas. It utilizes an ESP32 microcontroller and GPS module to continuously track the vehicle's position and compare it with predefined accident-prone zone coordinates stored in the system.

Whenever the vehicle approaches or enters a hazardous zone such as a school area, hospital zone, sharp curve, highway junction, or construction site, the system generates visual and audio alerts to notify the driver. These alerts help the driver become aware of the potential danger and take appropriate precautions. The system also monitors vehicle speed and compares it with the recommended speed limit for the identified zone.

If the vehicle exceeds the safe speed limit, an additional warning is generated to encourage the driver to slow down. This feature helps prevent accidents caused by over-speeding and improves driver response in critical situations. The warning mechanism is designed to be simple, effective, and easy to understand while driving.

The proposed system offers a cost-effective and reliable solution for enhancing road safety. By combining GPS-based location tracking, speed monitoring, and real-time alert generation, the system improves driver awareness, promotes safer driving behavior, and reduces the likelihood of accidents in high-risk areas. It can also serve as a foundation for future smart transportation and intelligent traffic management systems.

### **3.3 OBJECTIVES OF THE PROPOSED SYSTEM**

#### **3.3.1 Identification of Accident-Prone Zones**

The system identifies accident-prone zones using GPS coordinates and continuously tracks the vehicle's location. When the vehicle enters a hazardous area, it alerts the driver to improve awareness and ensure safer driving.

#### **3.3.2 Driver Alert System**

The system provides visual and audio alerts when the vehicle approaches an accident-prone zone. These timely warnings help drivers take necessary precautions, reduce speed, and improve overall road safety.

#### **3.3.3 Speed Monitoring**

The system continuously monitors the vehicle speed in accident-prone zones and compares it with the predefined safe speed limit. If the speed exceeds the limit, a warning is generated to encourage safer driving, reduce accident risks.

#### **3.3.4 Real-Time Location Tracking**

The system continuously tracks the vehicle's location using GPS technology. This enables accurate identification of accident-prone zones and ensures timely alerts are provided to the driver for improved road safety.

#### **3.3.5 Improvement of Road Safety**

The system aims to improve road safety by providing timely warnings and monitoring vehicle movement in accident-prone zones. This helps drivers make safer decisions, reduce accident risks, and ensure a safer transportation environment.

#### **3.3.6 Cost-Effective Implementation**

The proposed system is designed to be simple and cost-effective using readily available components such as ESP32 and GPS modules. This makes it suitable for installation in various vehicles while providing an efficient solution for improving road safety.



### **3.3.7 Future Scope and Expandability**

The system is designed with future expansion capabilities to support advanced road safety features. It can be integrated with IoT and smart transportation technologies for enhanced monitoring and accident prevention in the future.

### **3.3.8 Reduction of Accident Risks**

The proposed system helps reduce accident risks by providing timely alerts and monitoring vehicle movement in hazardous zones. This enables drivers to take preventive actions and promotes safer driving behavior on the road.

## **CHAPTER 4**

### **METHODOLOGY**

The methodology adopted for the proposed Accident Prone Zone Identification System is based on the integration of GPS technology, an ESP32 microcontroller, and a driver alert mechanism to enhance road safety. The system is designed to continuously monitor the real-time location of a vehicle and identify whether it is approaching or entering an accident-prone zone. A database containing the GPS coordinates of hazardous locations is stored in the system. The GPS module continuously receives satellite signals and provides the current geographical coordinates of the vehicle. These coordinates are then transmitted to the ESP32 microcontroller, which compares the vehicle's position with the predefined coordinates of accident-prone zones. If a match is detected or the vehicle enters a specified danger zone, the system recognizes the location as hazardous and initiates the warning process.

Once an accident-prone zone is identified, the ESP32 activates visual and audio warning devices to alert the driver. These alerts are intended to increase driver awareness and encourage precautionary measures before reaching the hazardous area. Along with location monitoring, the system also observes the vehicle's speed and compares it with the predefined safe speed limit for the respective zone. If the vehicle exceeds the recommended speed limit, an additional warning is generated to inform the driver about the unsafe driving condition. This dual-function approach of location-based identification and speed monitoring significantly improves the effectiveness of the safety system. The alert messages are generated instantly, allowing the driver to react quickly and reduce speed when necessary.

## Block Diagram

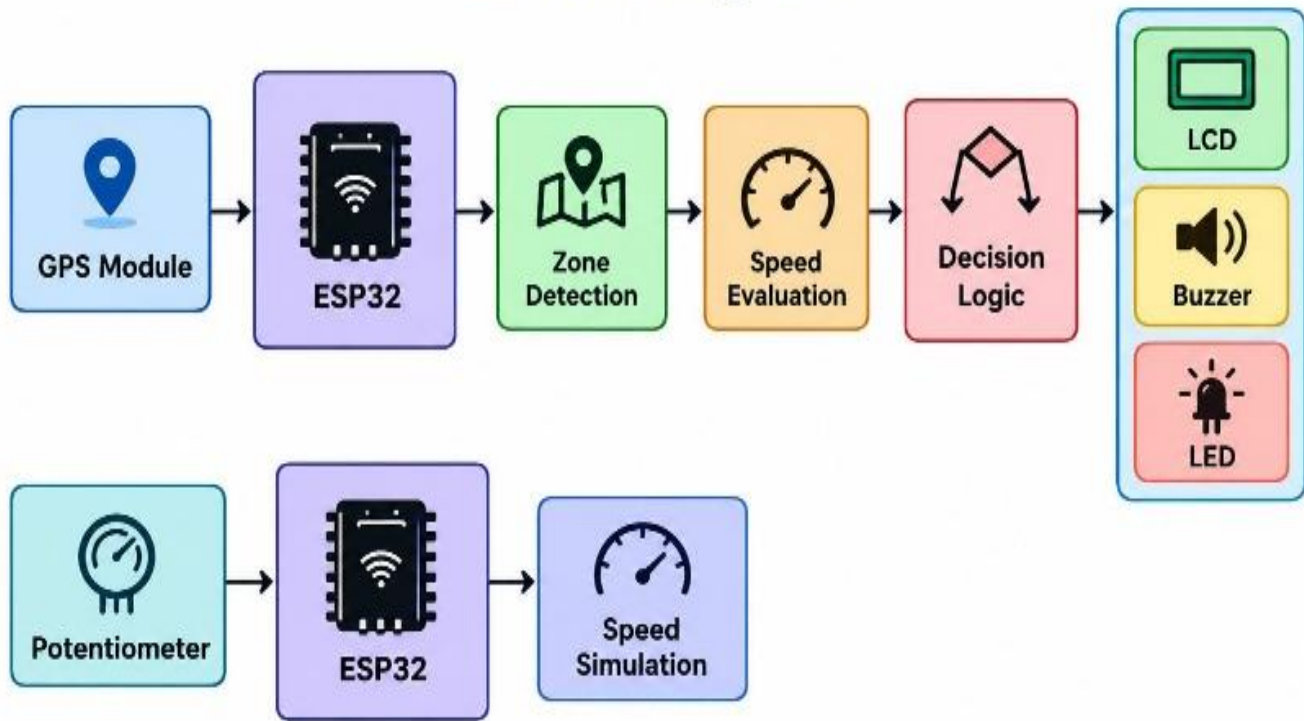


Fig 4.1 Block Diagram

The overall methodology focuses on preventing accidents rather than merely detecting them after occurrence. By providing real-time information and warnings, the system assists drivers in making safer decisions while traveling through critical locations such as school zones, hospital areas, sharp curves, highway intersections, bridges, and construction sites. The implementation of GPS-based tracking ensures accurate location identification, while the ESP32 controller provides efficient data processing and system control. The proposed methodology is simple, reliable, and cost-effective, making it suitable for installation in different types of vehicles. Through continuous monitoring, intelligent decision-making, and timely alert generation, the system contributes to reducing accident risks, improving driver awareness, and enhancing overall road transportation safety.

Furthermore, the proposed system operates automatically without requiring continuous user interaction. The GPS module continuously updates the vehicle's position, while the ESP32 controller processes the incoming data in real time. The system is designed to respond quickly and accurately, ensuring that warning messages are delivered before the vehicle reaches a hazardous location. This proactive approach helps drivers remain alert, avoid risky situations, and maintain safe driving behavior. The integration of location tracking, speed monitoring, and alert generation makes the system an effective solution for modern road safety applications.

The integration of location tracking, speed monitoring, and alert generation makes the system an effective solution for modern road safety applications. Through continuous monitoring and intelligent decision-making, the system can identify accident-prone zones and provide timely warnings to drivers. This helps reduce accident risks, improve driver awareness, and encourage safer driving practices. Overall, the proposed system provides a practical, reliable, and cost-effective approach to enhancing road transportation safety.

The proposed system provides a proactive approach to road safety by continuously monitoring the vehicle's location and identifying nearby accident-prone zones. When a hazardous area is detected, the system provides a timely warning through the display and alert mechanism, allowing the driver to take necessary precautions.

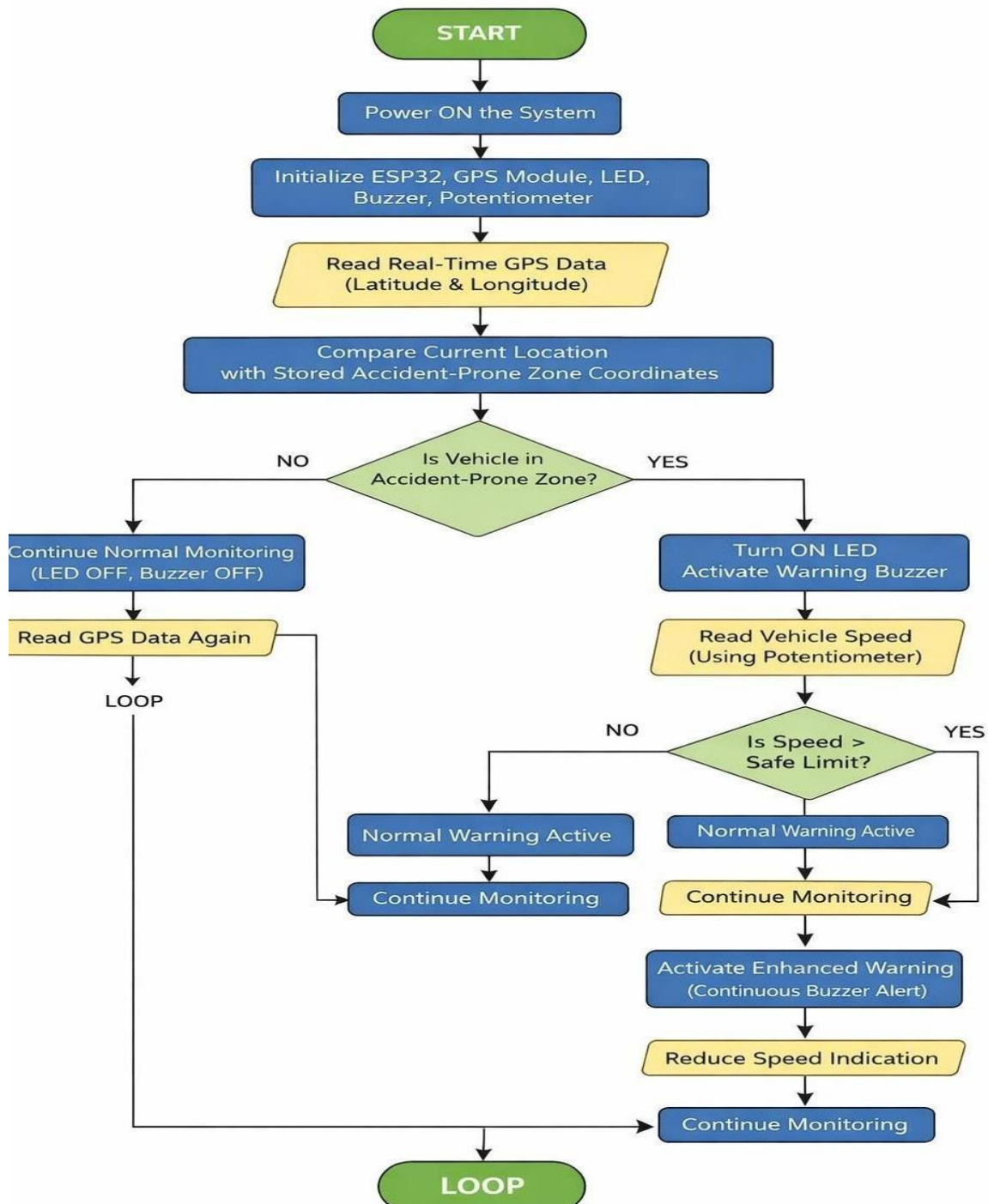


Fig. 4.2 Flowchart

## **CHAPTER 5**

### **HARDWARE AND COMPONENT SELECTION**

#### **5.1 ESP 32 MICROCONTROLLER**



Fig. 5.1 ESP 32 Microcontroller

The ESP32 is a powerful, low-cost microcontroller developed by Systems. It features a dual-core processor with integrated Wi-Fi and Bluetooth capabilities, making it suitable for real-time embedded applications. In this project, the ESP32 acts as the main control unit that processes GPS data, monitors speed values, and controls alert mechanisms.

The ESP32 receives latitude and longitude data from the GPS module through serial communication. It compares this data with predefined accident-prone zone coordinates stored in its memory. Based on this comparison and the vehicle speed value, the ESP32 decides whether to activate warning alerts. Its fast processing speed and multiple GPIO pins make it ideal for handling multiple peripherals simultaneously.

## 5.2 GPS MODULE (NEO-6M)

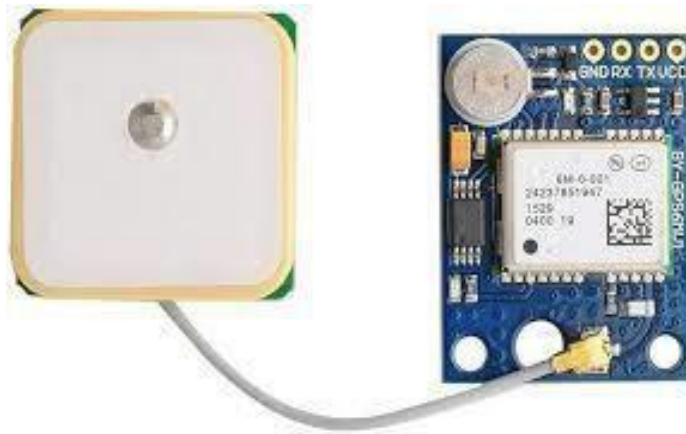


Fig. 5.2 GPS Module (NEO-6M)

The NEO-6M GPS module is used to determine the real-time geographical location of the vehicle. It receives signals from multiple GPS satellites and calculates accurate latitude and longitude values. The module provides continuous location updates, which are essential for identifying accident-prone zones.

In this system, the GPS module communicates with the ESP32 using UART communication. The location data obtained from the GPS is continuously compared with predefined coordinates to detect whether the vehicle has entered a hazardous zone. The GPS module ensures reliable outdoor performance and accurate positioning, making it suitable for vehicle-based applications.

### 5.3 BUZZER



Fig. 5.3 Buzzer

The buzzer is used as an audio alert device to warn the driver when the vehicle enters an accident-prone zone or exceeds the speed limit. It provides immediate auditory feedback, which is crucial for capturing the driver's attention without requiring visual focus. In system, the buzzer is activated by the ESP32 when specific conditions are met. A normal warning sound is generated when the vehicle enters a dangerous zone, enhanced sound is produced when over-speeding is detected. This ensures clear differentiation between normal and critical alerts.

### 5.4 LED INDICATOR



Fig. 5.4 LED Indicator



The LED serves as a visual alert to indicate the system status and warning conditions. When the vehicle enters an accident-prone zone, the LED turns ON to alert the driver. The LED remains OFF during normal driving conditions. Visual alerts are especially useful in noisy environments where the buzzer sound may not be clearly audible. The combination of LED and buzzer alerts improves system reliability and ensures better driver awareness.

### **5.5 POTENTIOMETER (SPEED SIMULATION UNIT)**



Fig. 5.5 Potentiometer (Speed Simulation Unit)

The potentiometer is used to simulate vehicle speed in the prototype model. It provides an analog voltage signal that varies with the rotation of the knob. This analog signal is read by the ESP32 and converted into a digital speed value. By adjusting the potentiometer, different speed conditions can be simulated, such as normal speed and over-speeding. This allows effective testing and validation of the speed monitoring and alert logic without requiring an actual vehicle speed sensor.

## 5.6 POWER SUPPLY UNIT



Fig. 5.6 Battery

The power supply provides the required voltage to all system components. The ESP32 typically operates at 3.3V or 5V, while the GPS module and other peripherals also require stable voltage for reliable operation. A regulated power source is used to ensure uninterrupted functioning of the system. Proper power management helps prevent system resets and improves overall reliability.

## 5.7 LCD DISPLAY



Fig.5.7 LCD Display

The LCD display shows the accident-prone zone status and alerts the user when a hazardous location is detected. It provides clear, real-time information about the identified accident-prone area for safer travel.

**CHAPTER 6**  
**COST ESTIMATION**

| <b>S.NO</b>       | <b>ITEM NAME</b>       | <b>NOS</b> | <b>PRICE</b> |
|-------------------|------------------------|------------|--------------|
| <b>1</b>          | ESP 32 Microcontroller | 1          | 400          |
| <b>2</b>          | GPS Module             | 1          | 350          |
| <b>3</b>          | Buzzer                 | 1          | 30           |
| <b>4</b>          | Potentiometer          | 1          | 150          |
| <b>5</b>          | LED Indicator          | 1          | 10           |
| <b>6</b>          | Battery                | 1          | 40           |
| <b>7</b>          | LCD Display            | 1          | 250          |
| <b>Total cost</b> |                        |            | <b>1230</b>  |

Table 6.1 Cost Estimation

# CHAPTER 7

## SYSTEM IMPLEMENTATION, OPERATION AND TESTING

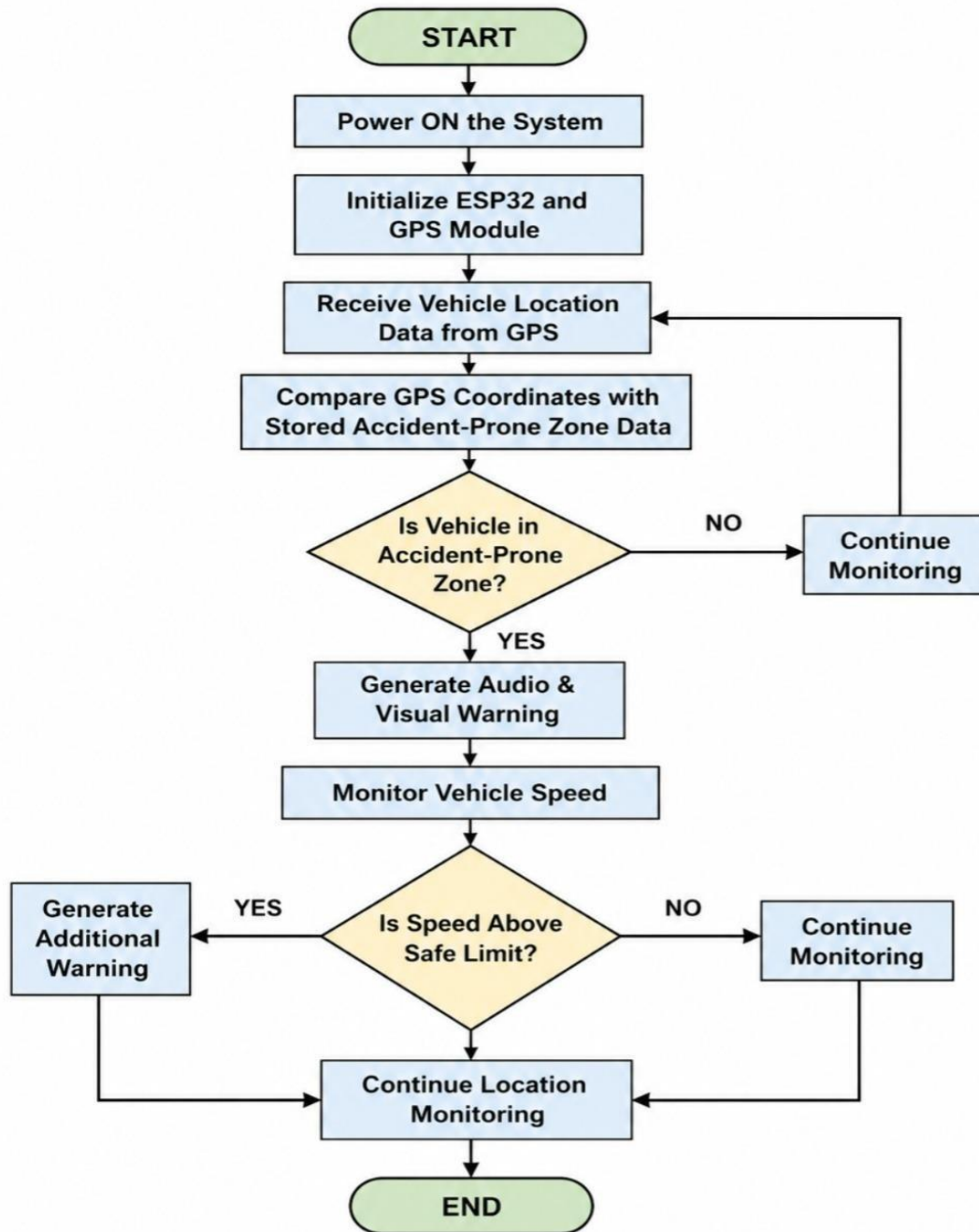


Fig. 7.1 System Operation

## 7.1 WORKING

- The working of the Accident Prone Zone Identification System begins when the system is powered on. The ESP32 microcontroller initializes all the connected components, including the GPS module, buzzer, and display unit. The GPS module continuously receives satellite signals and determines the real-time location of the vehicle. This location data is transmitted to the ESP32 for processing and analysis.
- The ESP32 compares the received GPS coordinates with the predefined coordinates of accident-prone zones stored in the system memory. If the vehicle is detected within or near a hazardous location, the system immediately generates visual and audio alerts to warn the driver. These alerts help the driver become aware of the danger and take necessary precautions while driving.
- In addition to location monitoring, the system also checks the vehicle speed in accident-prone zones. If the speed exceeds the predefined safe limit, an additional warning is generated through the buzzer and display unit. The system continuously repeats this process throughout the journey, ensuring real-time monitoring, improved driver awareness, and enhanced road safety.

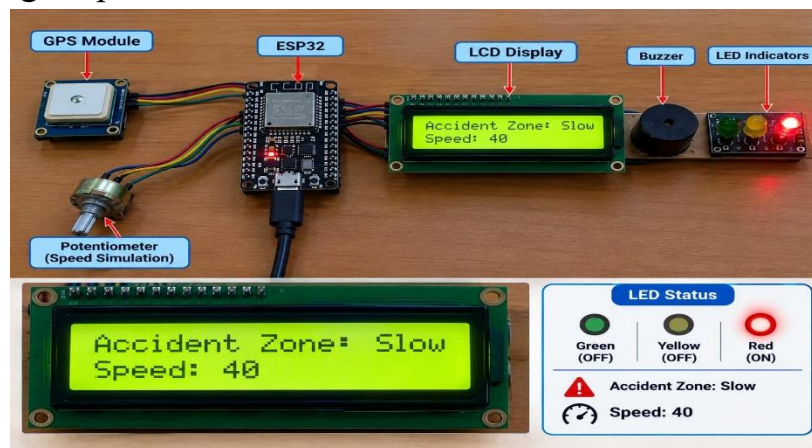


Fig. 7.2 Prototype Testing

### **7.1.1 GPS Data Acquisition**

The first step in the system operation is the acquisition of location data using the GPS module. The GPS module continuously receives signals from satellites and determines the real-time geographical coordinates of the vehicle. These coordinates are transmitted to the ESP32 microcontroller for further processing. Accurate location tracking enables the system to identify whether the vehicle is approaching or entering an accident-prone zone and ensures timely warning

### **7.1.2 Accident-Prone Zone Detection**

The ESP32 microcontroller compares the current GPS coordinates of the vehicle with the predefined coordinates of accident-prone zones stored in the system. When the vehicle approaches or enters a hazardous location, the system detects the zone and initiates the warning process, enabling drivers to take necessary precautions and improve road safety.

### **7.1.3 Driver Alert Generation**

When an accident-prone zone is detected, the system generates visual and audio alerts to warn the driver about the hazardous location. These alerts provide timely information, helping the driver reduce speed, remain cautious, and take necessary safety measures while traveling through the identified zone.

### **7.1.4 Speed Monitoring and Warning**

The system continuously monitors the vehicle speed while traveling through accident-prone zones. If the detected speed exceeds the predefined safe limit, the ESP32 generates an additional warning through the buzzer and display unit, prompting the driver to reduce speed and ensure safer driving conditions.

### **7.1.5 Driver Alert Mechanism**

The system continuously monitors the vehicle's location and speed throughout the journey. By repeatedly updating GPS data and checking for accident-prone zones, the system ensures timely alerts and maintains effective road safety monitoring at all times.

### **7.1.6 Improvement of Road Safety**

The primary outcome of the system operation is the improvement of road safety through real-time monitoring and timely warning generation. By alerting drivers about accident-prone zones and unsafe driving conditions, the system helps reduce accident risks, enhances driver awareness, and promotes safer transportation.

### **7.1.7 System Reliability**

The proposed system provides a reliable and efficient solution for identifying accident-prone zones and alerting drivers in real time. Its modular design allows future enhancements such as IoT connectivity, cloud-based data storage, and integration with smart transportation systems to further improve road safety and traffic management.

### **7.1.8 Overall System Performance**

The overall performance of the proposed system demonstrates its ability to accurately identify accident-prone zones and provide timely alerts to drivers. Through continuous GPS tracking, speed monitoring, and real-time warning generation, the system enhances driver awareness, reduces accident risks, and contributes to safer road transportation.

### **7.1.9 Results and Validation**

The Accident Prone Zone Identification and Prevention System was

successfully developed and tested using ESP32 and NEO-6M GPS. The system tracks vehicle location and provides audio and visual alerts when approaching an accident-prone zone, with the alert distance configured at approximately **30 metres**. It also monitors vehicle speed and gives an additional warning when the speed exceeds the **40 km/h** safe limit. The prototype demonstrates effective real-time hazard detection, improves driver awareness, and helps reduce accident risks with a calculated power consumption of **1.300 W**.

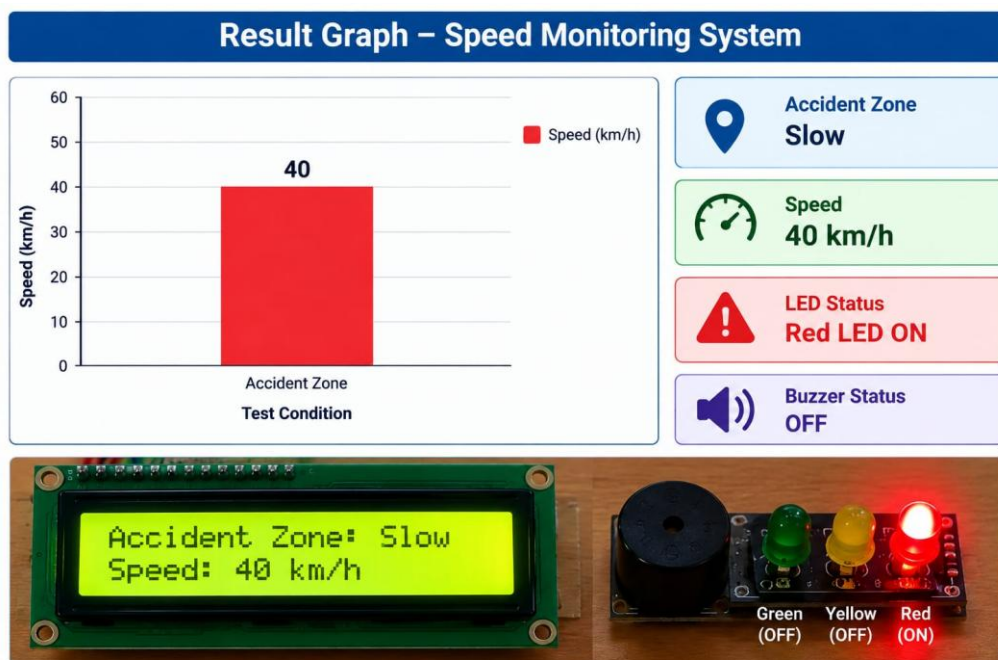


Fig. 7.3 Speed Monitoring System



## CHAPTER 8

### CONCLUSION

Road safety has become a critical concern due to the increasing number of vehicles and frequent accidents occurring in accident-prone zones such as sharp curves, school areas, hospital zones, highways, and construction sites. Many accidents are caused by over-speeding, lack of timely warnings, and insufficient driver awareness. To address these issues, this project successfully designed and implemented an Accident Prone Zone Identification and Prevention System using GPS and ESP32. The proposed system effectively utilizes GPS technology to continuously track the real-time location of a vehicle and compare it with predefined accident-prone zone coordinates. When the vehicle enters a hazardous zone, the system immediately alerts the driver using visual and audio indicators. This early warning mechanism helps drivers take preventive action and reduces the likelihood of accidents. The integration of speed monitoring further enhances safety by identifying over-speed conditions within critical zones and generating enhanced warnings.

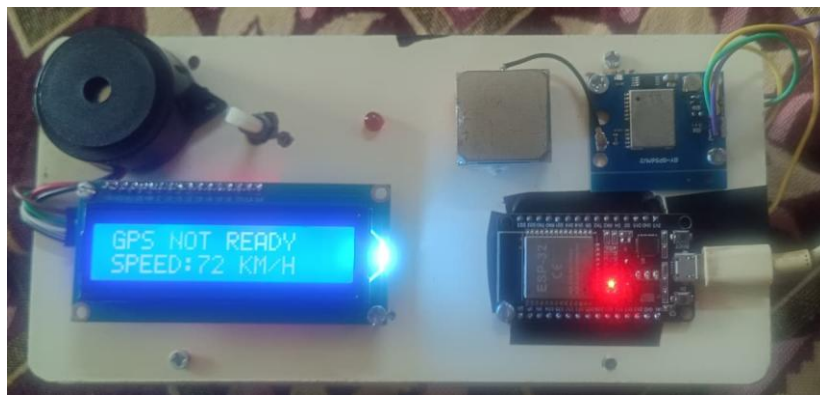


Fig 8.1 Prototype Output

## **CHAPTER 9**

### **ADVANTAGES, APPLICATIONS AND FUTURE SCOPE**

#### **9.1 ADVANTAGES**

- **Early Warning:** Alerts the driver before or when approaching an accident-prone zone.
- **Adjustable Alert Distance:** The warning distance can be configured according to the road condition and application requirements.
- **Real-Time GPS Tracking:** Continuously monitors the vehicle's geographical location.
- **Speed Monitoring:** Detects over-speeding in hazardous zones and provides an additional warning.
- **Audio and Visual Alerts:** The combination of buzzer and LED makes the warning easy to notice.
- **Improves Driver Awareness:** Helps drivers identify hazardous locations that may not be immediately visible.
- **Cost-Effective:** Uses readily available components such as ESP32 and NEO-6M GPS.
- **Easy to Implement:** The system has a simple architecture and can be installed in different types of vehicles.
- **Low Power Consumption:** The prototype has a calculated power consumption of approximately 1.300 W.
- **Expandable System:** The system can be further developed using IoT, cloud platforms, GSM/4G/5G, AI, and smart transportation technologies.

## 9.2 APPLICATION

- Cars and Passenger Vehicles – Provides drivers with warnings near dangerous locations.
- Buses and Public Transportation – Helps bus drivers identify accident-prone areas along their routes.
- School Zones – Provides warnings near schools where reduced speed and increased attention are required.
- Hospital Zones – Helps drivers maintain safe speeds near hospitals and sensitive areas.
- Highways and Highway Junctions – Warns drivers about identified high-risk highway locations.
- Sharp Curves and Dangerous Turns – Provides advance warnings before entering critical road sections.
- Construction Zones – Alerts drivers near road construction and temporary hazardous areas.
- Bridges and Critical Road Sections – Helps improve driver awareness in locations requiring additional caution.
- Fleet Management Systems – Can be integrated into commercial vehicle fleets for location and speed monitoring.
- Smart Transportation Systems – Can be expanded with IoT, cloud connectivity, and intelligent traffic-management technologies.
- Emergency Response Systems – Future versions can communicate hazardous-location information to emergency response systems.
- Mining and Industrial Safety – With further development, the concept can be adapted for worker and vehicle safety in hazardous environments.

### **9.3 LIMITATIONS**

- **Data Dependency** – The accuracy of the system depends heavily on the quality, quantity, and completeness of accident data.
- **Limited Historical Data** – If sufficient past accident records are unavailable, the system may not accurately identify high-risk zones.
- **Changing Road Conditions** – Road construction, traffic diversions, weather conditions, and changes in road design can affect accident-prone areas over time.
- **GPS Accuracy Issues** – Incorrect or imprecise location data can lead to errors in identifying the exact accident-prone location.
- **Real-Time Data Limitations** – Continuous monitoring requires reliable sensors, GPS, internet connectivity, and other infrastructure.
- **Human Factors** – Driver behavior, speeding, distracted driving, fatigue, and violations are difficult to measure accurately and may not be fully considered.
- **Environmental Factors** – Fog, rain, poor lighting, and road surface conditions can suddenly increase accident risk but may be difficult to predict.
- **False Predictions** – The system may classify a location as high-risk even when accidents are unlikely, or fail to identify a newly emerging dangerous location.

### **9.4 FUTURE SCOPE**

- **Real-Time Accident Prediction** – The system can be enhanced to predict accident risks in real time using live traffic, weather, road, and vehicle data.
- **AI and Machine Learning** – Advanced machine-learning and deep-learning algorithms can improve the accuracy of identifying and predicting accident-prone zones.
- **IoT Integration** – IoT-based sensors can be installed on roads and vehicles to continuously collect information about speed, traffic density, road conditions, and accidents.

- Smart City Integration – The system can be integrated with smart-city traffic management systems to monitor and manage dangerous road sections.
- GPS and Mobile Application – A mobile application can provide drivers with alerts about nearby accident-prone zones and suggest safer routes.
- Weather-Based Risk Analysis – Weather information such as heavy rain, fog, and poor visibility can be combined with accident data to improve risk prediction.
- Automatic Emergency Response – In the event of an accident, the system could automatically notify nearby hospitals, police, and emergency services with the accident location.
- Predictive Traffic Management – Authorities can use the system's predictions to implement speed restrictions, traffic diversions, improved lighting, or additional road signs in high-risk areas.
- Integration with Smart Vehicles – Future vehicles can receive warnings directly from the system and provide drivers with collision-risk alerts.
- Advanced Mapping and Visualization – GIS and interactive maps can be used to display accident hotspots and analyze accident patterns more effectively.
- Continuous Data Updating – The system can continuously learn from newly collected accident and traffic data, making predictions more accurate over time.
- Expansion to Large-Scale Networks – The system can be expanded from individual roads or cities to cover entire districts, states, or national highway networks.

## CHAPTER 10

### APPENDICES

#### 10.1 APPENDIX – 1 (PHOTOGRAPHY)

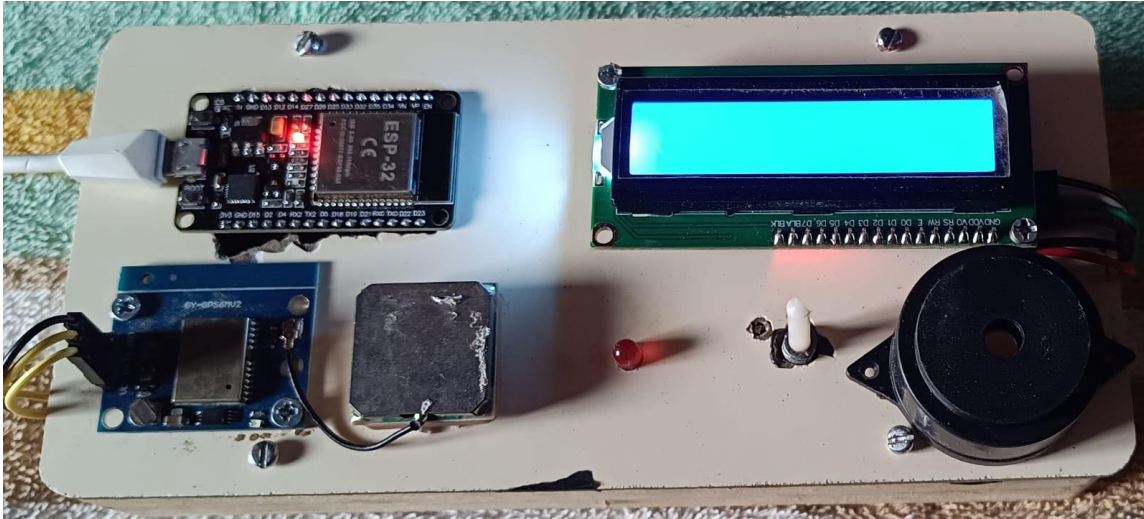


Fig. 10.1 Prototype Photograph

#### 10.2 APPENDIX – 2 (CODE)

```
#include <Wire.h>
#include <LiquidCrystal_I2C.h>
#include <TinyGPSPlus.h>
#define POT_PIN 34
#define LED_PIN 26
#define BUZZER_PIN 25
#define GPS_RX 18
#define GPS_TX 19
#define SDA_PIN 21
```

```

#define SCL_PIN 22

LiquidCrystal_I2C lcd(0x27, 16, 2);
TinyGPSPlus gps;
HardwareSerial GPS_Serial(1);

double accidentLat = 11.6662747;
double accidentLng = 78.1786040;
double warningDistance = 500.0;
double accidentRadius = 100.0;
int safeSpeed = 40;
unsigned long lastLCDUpdate = 0;
unsigned long lastBlink = 0;
bool ledState = false;

void setup()
{
  Serial.begin(115200);
  pinMode(POT_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  pinMode(BUZZER_PIN, OUTPUT);
  digitalWrite(LED_PIN, LOW);
  digitalWrite(BUZZER_PIN, LOW);
  Wire.begin(SDA_PIN, SCL_PIN);
  lcd.init();
  lcd.backlight();
  lcd.clear();

```

```

    lcd.setCursor(0, 0);
    lcd.print("ACCIDENT ZONE");
    lcd.setCursor(0, 1);
    lcd.print("SYSTEM START");
    delay(2000);
    GPS_Serial.begin(9600, SERIAL_8N1, GPS_RX,
GPS_TX);
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("GPS SEARCHING");
    lcd.setCursor(0, 1);
    lcd.print("PLEASE WAIT");
}
void loop()
{
    while (GPS_Serial.available() > 0)
    {
        gps.encode(GPS_Serial.read());
    }
    int potValue = analogRead(POT_PIN);
    potValue = 4095 - potValue;
    int speed = map(potValue, 0, 4095, 0, 120);
    speed = constrain(speed, 0, 120);
    if (gps.location.isValid())

```



```

{
    double currentLat = gps.location.lat();
    double currentLng = gps.location.lng();
    double distance = TinyGPSPlus::distanceBetween(
        currentLat,
        currentLng,
        accidentLat,
        accidentLng
    );
    if (distance <= accidentRadius)
    {
        if (speed > safeSpeed)
        {
            digitalWrite(BUZZER_PIN, HIGH);
            if (millis() - lastBlink >= 300)
            {
                lastBlink = millis();
                ledState = !ledState;
                digitalWrite(LED_PIN, ledState);
            }

            if (millis() - lastLCDUpdate >= 1000)
            {
                lastLCDUpdate = millis();

```

```

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("ACCIDENT ZONE");
    lcd.setCursor(0, 1);
    lcd.print("SLOW DOWN");
}
}
else
{
    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);
    ledState = false;

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("ACCIDENT ZONE");
    lcd.setCursor(0, 1);
    lcd.print("SPEED:");
    lcd.print(speed);
    lcd.print(" KM/H");
}
}
else if (distance <= warningDistance)

```

```

{
  if (speed > safeSpeed)
  {
    digitalWrite(BUZZER_PIN, HIGH);

    if (millis() - lastBlink >= 300)
    {
      lastBlink = millis();
      ledState = !ledState;
      digitalWrite(LED_PIN, ledState);
    }

    if (millis() - lastLCDUpdate >= 1000)
    {
      lastLCDUpdate = millis();

      lcd.clear();
      lcd.setCursor(0, 0);
      lcd.print("ZONE AHEAD");
      lcd.setCursor(0, 1);
      lcd.print("SLOW DOWN");
    }
  }
  else

```

```

{
    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);
    ledState = false;

    if (millis() - lastLCDUpdate >= 1000)
    {
        lastLCDUpdate = millis();

        lcd.clear();
        lcd.setCursor(0, 0);
        lcd.print("ZONE AHEAD");
        lcd.setCursor(0, 1);
        lcd.print("SPEED:");
        lcd.print(speed);
        lcd.print(" KM/H");
    }
}
}
else
{
    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);
    ledState = false;

```

```

if (millis() - lastLCDUpdate >= 1000)
{
    lastLCDUpdate = millis();

    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("SAFE ZONE");
    lcd.setCursor(0, 1);
    lcd.print("SPEED:");
    lcd.print(speed);
    lcd.print(" KM/H");
}
}
else
{
    digitalWrite(LED_PIN, LOW);
    digitalWrite(BUZZER_PIN, LOW);
    ledState = false;

    if (millis() - lastLCDUpdate >= 1000)
    {
        lastLCDUpdate = millis();
    }
}

```

```
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print("GPS NOT READY");  
    lcd.setCursor(0, 1);  
    lcd.print("SPEED:");  
    lcd.print(speed);  
    lcd.print(" KM/H");  
}  
}  
  
delay(50);  
}
```

## REFERENCES

- 1.S. K. Prasad, P. Mahalakshmi, and A. John, “GPS and GSM Based Accident Detection and Alert System,” *International Journal of Engineering Research & Technology (IJERT)*, vol. 3, no. 4, pp. 456–459, 2014.
- 2.R. Kumar and M. Singh, “Intelligent Speed Control System Using GPS for Accident Prevention,” *International Journal of Advanced Research in Computer Engineering & Technology (IJARCET)*, vol. 4, no. 6, pp. 2341–2345, 2015.
- 3.A. Sharma and V. Patel, “IoT Based Smart Vehicle Monitoring and Accident Prevention System,” *International Journal of Scientific Research in Computer Science, Engineering and Information Technology*, vol. 5, no. 2, pp. 112–118, 2019.
- 4.K. Ramesh and S. Anitha, “Accident Prone Zone Detection Using GPS and Embedded Systems,” *International Journal of Emerging Technology and Advanced Engineering*, vol. 6, no. 3, pp. 89–94, 2016.
- 5.P. N. Rao and R. Karthik, “Vehicle Speed Monitoring and Warning System Using Microcontroller,” *International Journal of Electronics and Communication Engineering*, vol. 8, no. 1, pp. 21–26, 2017.
- 6.Espressif Systems, *ESP32 Technical Reference Manual*, Espressif Systems Inc., 2023. U-blox AG, *NEO-6 GPS Modules Data Sheet*, Switzerland, 2020.

